Project 2 - ViGoEmotions 2026: Fine-Grained Multi-Label Vietnamese Emotion Recognition

Dataset & Problem

ViGoEmotions is a Vietnamese multi-label emotion benchmark dataset presented at the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026) [1]. It contains 20,664 social-media comments manually annotated with 27 fine-grained emotion categories (as well as an explicit neutral state mapping):

Property	Value
Size	20,664 user comments
Label space	27 fine-grained emotion categories + neutral state
Task type	Multi-label text classification
Language	Vietnamese (informal social-media discourse)

A single comment may simultaneously express multiple emotions (multi-label classification). Unlike single-label classification where categories are mutually exclusive, a multi-label model produces an independent binary decision for each category. Consequently, exact-match subset accuracy (which penalizes any partial discrepancy across all 27 labels) is overly stringent; the primary evaluation metrics are Macro-averaged F1-score (Macro-F1) and Micro-averaged F1-score (Micro-F1) - **not subset accuracy**.

Dataset-use rule: Adhere strictly to the dataset research license (restricted to non-commercial academic research and education). Do not redistribute raw comment text in any public repository. Student repositories may host source code, data loading scripts, split index files, derived statistical aggregates, and anonymized non-reconstructable evaluation metrics.

Resource	Link
Original paper (EACL 2026)	https://doi.org/10.18653/v1/2026.eacl-long.129
Dataset repository	https://huggingface.co/datasets/uitnlp/vigoemotions

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Requirements

Midterm

A. Dataset Analysis

Report:

- total sample count and the complete 27-label inventory;
- label frequency distribution, identifying head (frequent), mid-tier, and tail (rare) emotion labels;
- average number of positive labels per instance, known as label cardinality distribution ($|Y|$);
- label co-occurrence patterns and pairwise correlation matrix;
- text length distributions measured in both whitespace-delimited tokens and characters;
- qualitative review of Vietnamese informal text characteristics: internet slang, teencode (Vietnamese youth shorthand, character substitutions, and phonetic abbreviations), emojis, and non-standard syntax;
- stratified multi-label train / validation / test partition protocol.

B. Required Models

Implement and benchmark:

1. Term Frequency - Inverse Document Frequency (TF-IDF) representation paired with One-vs-Rest (OvR) Logistic Regression;
2. Recurrent neural network baseline: Bidirectional Long Short-Term Memory (BiLSTM) [7] or Gated Recurrent Unit (GRU) [8];
3. Pretrained transformer fine-tuning: PhoBERT [9] (a monolingual RoBERTa-based architecture pretrained on Vietnamese text) equipped with an independent sigmoid classification head and Binary Cross-Entropy (BCE) loss:

$$\mathcal{L} = -\frac{1}{C}\sum_{c=1}^{C}\left[y_c \log p_c + (1-y_c)\log(1-p_c)\right].$$

C. Required Metrics

- Macro-averaged F1-score (Macro-F1) and Micro-averaged F1-score (Micro-F1);
- per-label F1-scores across all 27 categories;
- Hamming Loss (the fraction of incorrectly predicted label-instance pairs across all labels);
- ranking performance: Label Ranking Average Precision (LRAP) or Mean Average Precision (mAP);
- model parameter count and wall-clock training time.

D. Midterm Questions

1. Which emotion labels exhibit the highest and lowest predictive F1-scores?
2. How strongly does model performance correlate with training label frequency?
3. Which emotion pairs co-occur most frequently, and does the baseline model preserve these dependencies?
4. On which text styles (short comments, emoji-heavy text, teencode) does PhoBERT most significantly outperform the recurrent baseline?
5. How sensitive are baseline classification outputs to the arbitrary default decision threshold of 0.5?

E. Midterm Deliverables

- presentation slides;
- executable Jupyter notebook / Python scripts;
- dataset-analysis summary tables and distribution plots;
- baseline experimental results table (all required metrics per model);
- detailed error analysis on edge cases;
- proposed research question for the Final evaluation;
- project README documentation.

Final

Required Research Question (RQ1):

> Is a single uniform decision threshold ($\tau = 0.5$) optimal across all 27 emotion labels in the presence of severe class imbalance?

Employ the highest-performing neural architecture established during the Midterm. Do not retrain the network backbone when modifying decision thresholds.

A. Compare Three Threshold Decoding Strategies

Train a single model on the training set. Optimize thresholds strictly on validation set predictions (D_{val}), then evaluate exactly once on the held-out test set [5], [6]:

Strategy	Formal Definition	Description
Fixed	$\tau = 0.5$	Standard default decision threshold applied uniformly to all labels $c \in \{1, \ldots, C\}$
Global tuned	$\tau^* = \arg\max_\tau F_1^{macro}(D_{val}; \tau)$	Single scalar threshold optimized to maximize Macro-F1 across all labels on validation data
Per-label tuned	$\tau_c^* = \arg\max_{\tau_c} F_{1,c}(D_{val}; \tau_c)$	Independent decision threshold τ_c tuned individually for each emotion class c on validation data

Freeze all calibrated thresholds prior to performing test-set inference.

B. Required Analysis

Stratify emotion classes into head (frequent), mid-tier, and tail (rare) groups using a clearly documented frequency threshold defined prior to testing. Report:

- Macro-F1, Micro-F1, per-label F1, and Hamming Loss across all three decoding strategies;
- performance gains or losses broken down by label frequency tier (head / mid / tail);
- performance breakdown stratified by label cardinality ($|Y| = 1, 2, 3, \geq 4$);
- calibrated threshold profile τ_c^* across the 27 emotion classes;
- validation-to-test generalization gap ($\Delta F_1 = F_1^{val} - F_1^{test}$) to identify potential threshold overfitting.

C. Required Error Analysis

Conduct an empirical audit on at least:

- 20 false-negative predictions (missed emotions);
- 20 false-positive predictions (spurious detections);
- complex multi-emotion comments with high label cardinality;
- systematically confused or mutually co-occurring emotion categories.

D. Final Deliverables

- conference-style technical report;
- reproducible Jupyter notebook / Python codebase;
- fixed split index files;
- hyperparameter configurations and training specifications;
- saved validation and test prediction probabilities (`.npz` array or CSV file with columns: `sample_id`, `true_label_vector`, `probability_vector`, `pred_fixed`, `pred_global`, `pred_per_label`, `label_cardinality`);
- programmatic plots and comparison tables;
- project README with execution guidelines.

Optional Extensions Attempt these extensions only after RQ1 has been fully completed and validated.

Extension 1 - Dependency-Aware Multi-Label Modeling (RQ2)

Compare the independent sigmoid baseline against architectures designed to model inter-label dependencies [2], [4]:

- Classifier Chains (CC) [3];
- label-correlation graph or Graph Convolutional Network (GCN) constructed exclusively from training-set label co-occurrences;
- lightweight label-attention mechanism.

Report performance deltas separately across head, mid, and tail labels, as well as by label cardinality.

Extension 2 - Imbalance-Aware Loss Functions (RQ3)

Evaluate whether Focal Loss [10] or Class-Balanced Loss [11] improves tail-label F1-score relative to standard Binary Cross-Entropy (BCE).

Extension 3 - Social-Media Text Noise Robustness (RQ4)

Benchmark model resilience against common Vietnamese online text corruptions: diacritic removal (unaccented Vietnamese), simulated typing errors (QWERTY / Telex keyboard layout noise), punctuation removal, or teencode word substitution.

Extension 4 - Full Paper Track: Joint Architecture and Threshold Optimization (RQ5)

Integrate dependency-aware representation modeling, per-label threshold calibration, and granular frequency stratification across at least three random seeds. Intended for student teams selected for academic publication.

Reproducibility Rules

- Under no circumstances should decision thresholds be calibrated on test-set data.
- Do not construct label-correlation graphs or co-occurrence matrices using test labels.
- Persist continuous prediction probabilities prior to threshold binarization.
- Fix all pseudo-random seeds and document all text normalization and tokenization steps.
- Record the exact pretrained model checkpoint names and versions.
- For paper-track studies, execute experiments across at least three random seeds and report standard deviations.

References

[1] T. Q. Hung, P. T. Nam, S. T. Luu, and K. V. Nguyen, "ViGoEmotions: A Benchmark Dataset For Fine-grained Emotion Detection on Vietnamese Texts," in *Proc. 19th Conf. Eur. Chapter Assoc. Comput. Linguistics (EACL)*, vol. 1, 2026, pp. 2805-2831, doi: 10.18653/v1/2026.eacl-long.129.

[2] C. Huang, A. Trabelsi, X. Qin, N. Farruque, L. Mou, and O. Zaïane, "Seq2Emo: A Sequence to Multi-Label Emotion Classification Model," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Hum. Lang. Technol. (NAACL-HLT)*, 2021, pp. 4717-4724, doi: 10.18653/v1/2021.naacl-main.375.

[3] D. J. W. Touw and M. Van de Velden, "Classifier chain networks for multi-label classification," *Expert Systems with Applications*, vol. 286, p. 128048, Aug. 2025, doi: 10.1016/j.eswa.2025.128048.

[4] C. Huang et al., "Multi-label emotion classification based on transfer information and external knowledge," *Knowledge-Based Systems*, vol. 327, p. 114089, Oct. 2025, doi: 10.1016/j.knosys.2025.114089.

[5] R. Alotaibi and P. Flach, "Multi-label thresholding for cost-sensitive classification," *Neurocomputing*, vol. 436, pp. 232-247, May 2021, doi: 10.1016/j.neucom.2020.12.004.

[6] I. Pillai, G. Fumera, and F. Roli, "Threshold optimisation for multi-label classifiers," *Pattern Recognition*, vol. 46, no. 7, pp. 2055-2065, Jul. 2013, doi: 10.1016/j.patcog.2013.01.012.

[7] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional LSTM and other neural network architectures," *Neural Networks*, vol. 18, no. 5-6, pp. 602-610, 2005, doi: 10.1016/j.neunet.2005.06.042.

[8] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1724-1734, doi: 10.3115/v1/D14-1179.

[9] D. Q. Nguyen and A. T. Nguyen, "PhoBERT: Pre-trained language models for Vietnamese," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1037-1042, doi: 10.18653/v1/2020.findings-emnlp.92.

[10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2980-2988, doi: 10.1109/ICCV.2017.324.

[11] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 9268-9277, doi: 10.1109/CVPR.2019.00949.